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Customer Stories Using Omnissa Horizon's REST API

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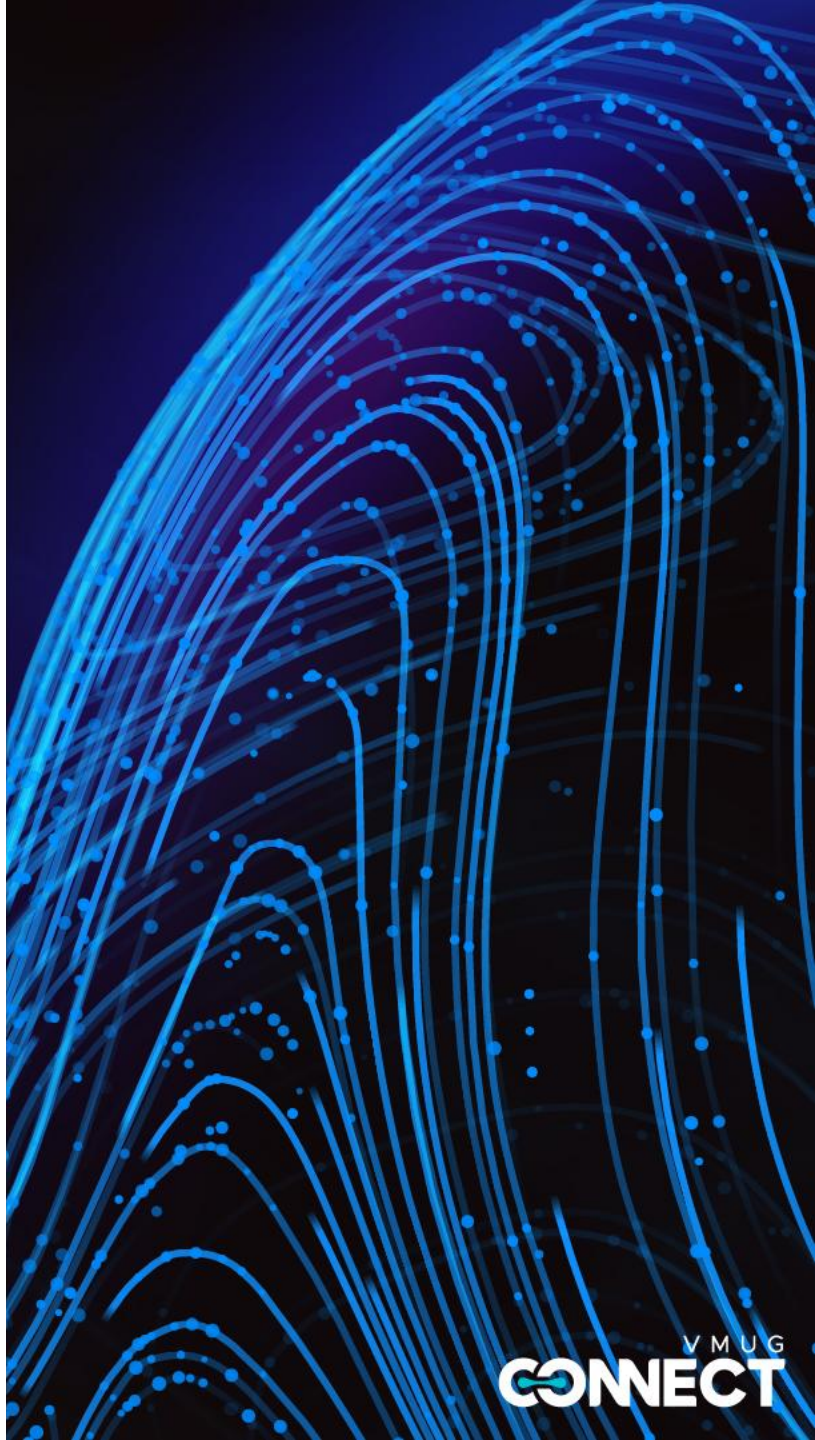
Systems Engineer – Minnesota State Colleges and Universities

Who am I?

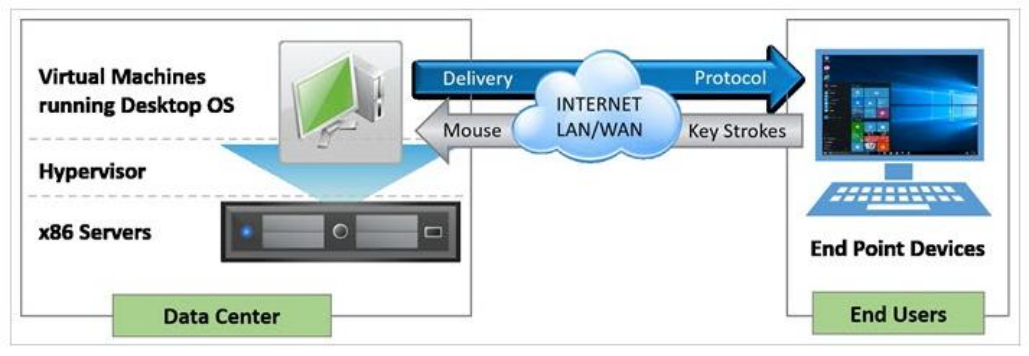
- Systems Engineer for Minnesota State (26y)
- Server Virtualization since 2006
- Desktop Virtualization since 2009
- VMware vExpert x 12
- Omnissa Tech Insider / EUC Champ x11
- VMUG Leader in Minneapolis
- Former VMUG Vice President / Director
- VCP-DCV/DT/VCF Admin/VCF Architect
- Rabid vCommunity Advocate
- Blog at www.vmatt.net

Let’s chat about VDI

- What is VDI?
- VDI at the Minnesota State System Office
- Current Architecture / Strategy
- Opportunities for Automation
- Omnissa Horizon REST API
- Use-cases at Minnesota State
- Live Demo



What is VDI?



- Virtual Desktop Infrastucture
  - Physical compute / storage / network infrastructure
  - A Hypervisor to create and manage virtual desktops
  - Virtual Machines that provide the OS for the user (Windows or Linux)
  - Connection Brokers / Gateways to Authenticate/Authorize users and allow access
  - Load Balancers (optional)
  - A client installed on an endpoint (optional if there is a web client)
  - Display Protocol that allows streaming to the desktop from your client device
  - Other components may include Active Directory, DHCP, KMS, File Servers, Persona Software
  - Antivirus / Antimalware

- VDI is typically on-premises.
- VDI in the cloud is usually referred to as DaaS (Desktop as a Service)

VDI at the Minnesota State System Office

- Virtual Desktop Infrastucture components
  - Physical compute / storage / network infrastructure – Dell MX Blades, HPE Alletra Storage, Cisco/Palo Alto
  - A Hypervisor to create and manage virtual desktops – VMware vSphere 8 (VVF for VDI)
  - Virtual Machines that provide the OS for the user – Windows 11
  - Connection Brokers / Gateways – Omnissa Horizon Connection Servers and Unified Access Gateways
  - Load Balancers – Citrix Netscaler VPX
  - A client installed on an endpoint – Mix of Omnissa Horizon Client and Web Client running
  - Display Protocol that allows streaming to the desktop from your client device – Allow users the choice of Blast, PCoIP or RDP, and the web client.
  - Other components may include Active Directory, DHCP, KMS, File Servers, Persona Software – Mostly Microsoft products here, and Liquidware Profile Unity for persona management.
  - Antivirus / Antimalware – Microsoft Defender for Endpoint

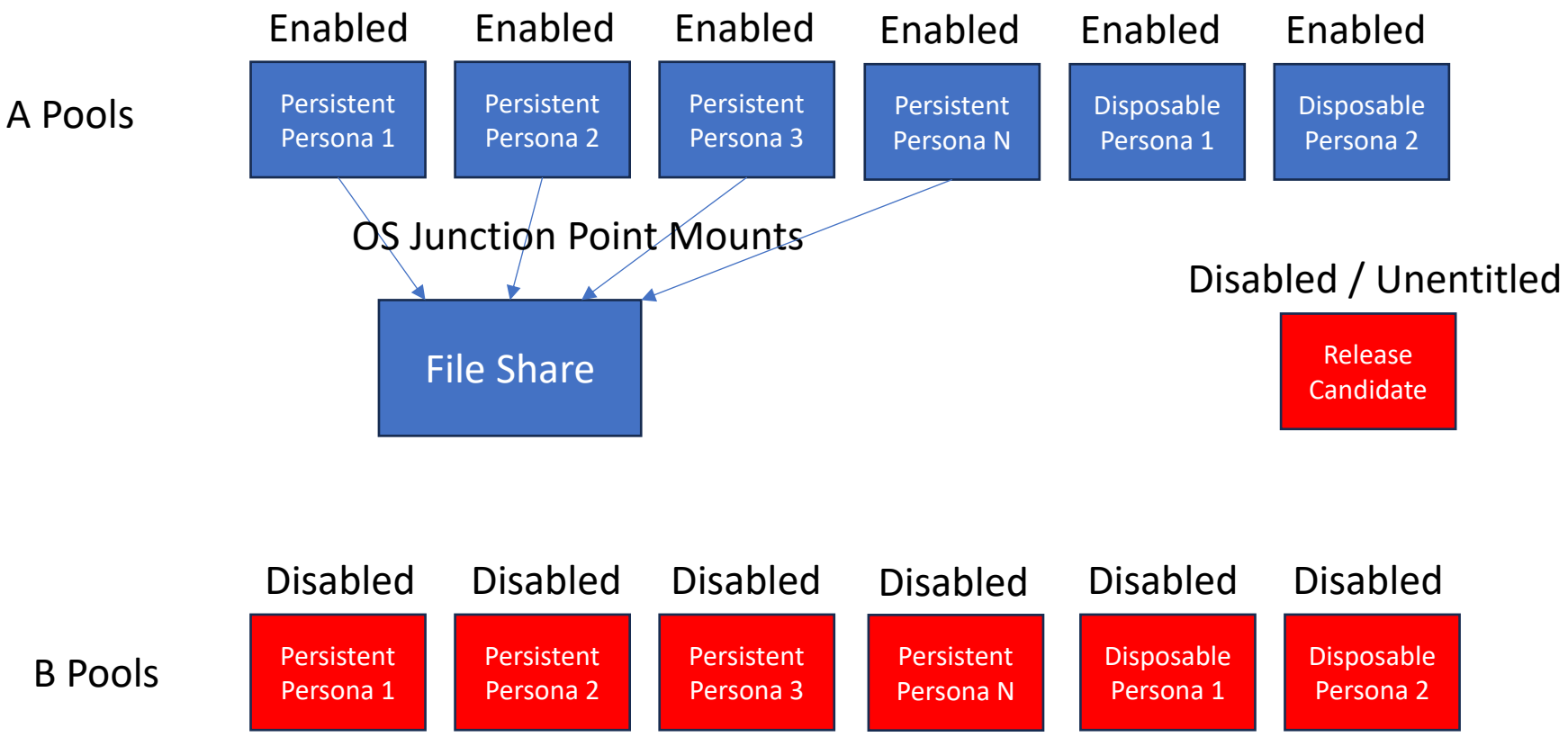
Current Architecture and Strategy

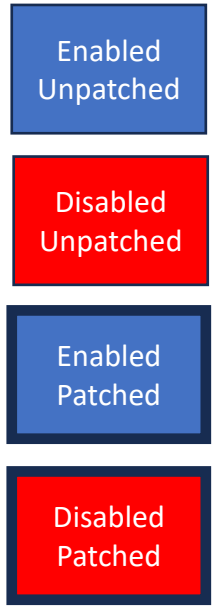
- Approximately 500 customer users
  - Range from daily, all-day use, to sporadic for temporary remote access
  - System Office and 6 of our 54 campuses
  - 80% Non-Persistent, Floating with a persistent persona
  - 20% Non-Persistent, Floating with a disposable persona
  - Redundant Gateways and Connection Servers – all load balanced with Netscaler

Current Architecture and Strategy

- Campuses have between 1 and 4 network security zones, each with a desktop pool
- Each pool contains a MAX limit between 20 and 70 machines with a minimum/spare machines set to 10% of the MAX
- Each pool deployment has 2 pools (A side and B side)
- Moves from A to B to A after each in-band or out-of-band

Current Architecture and Strategy (Operations)





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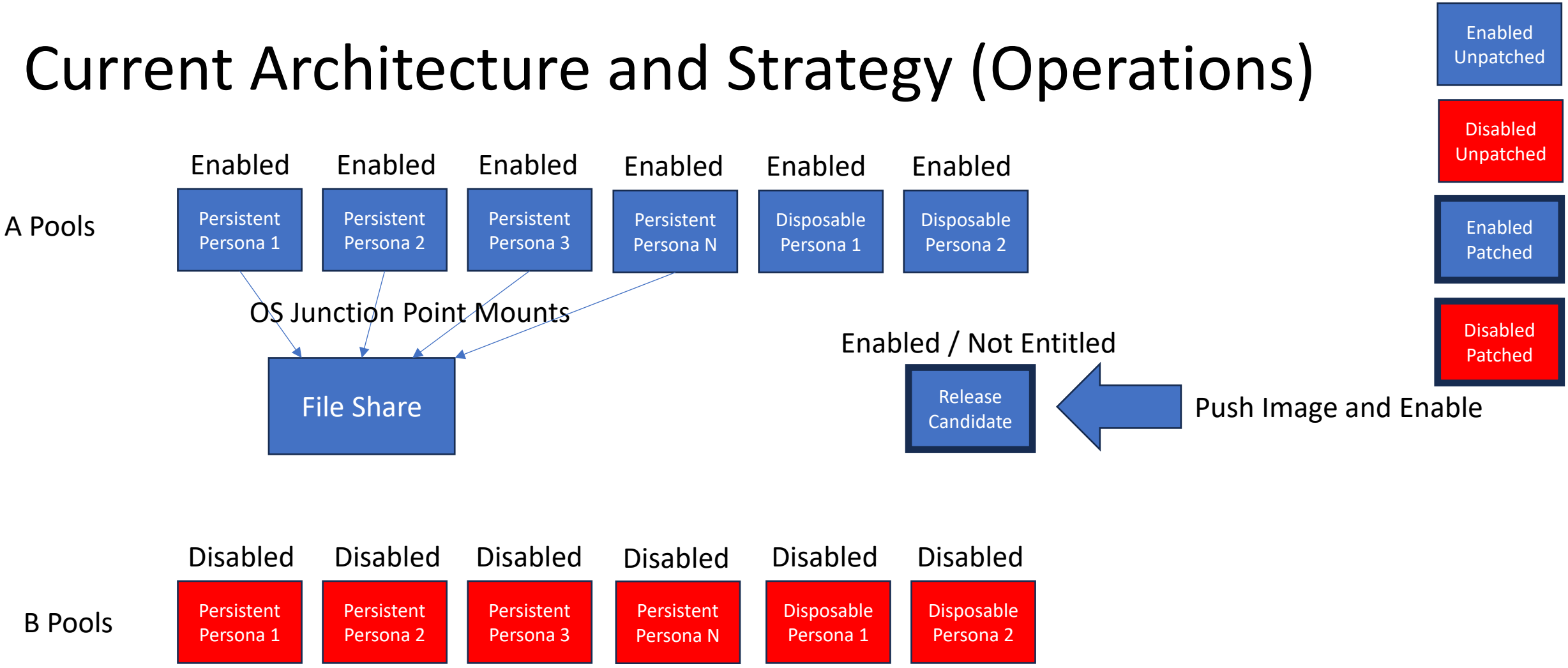

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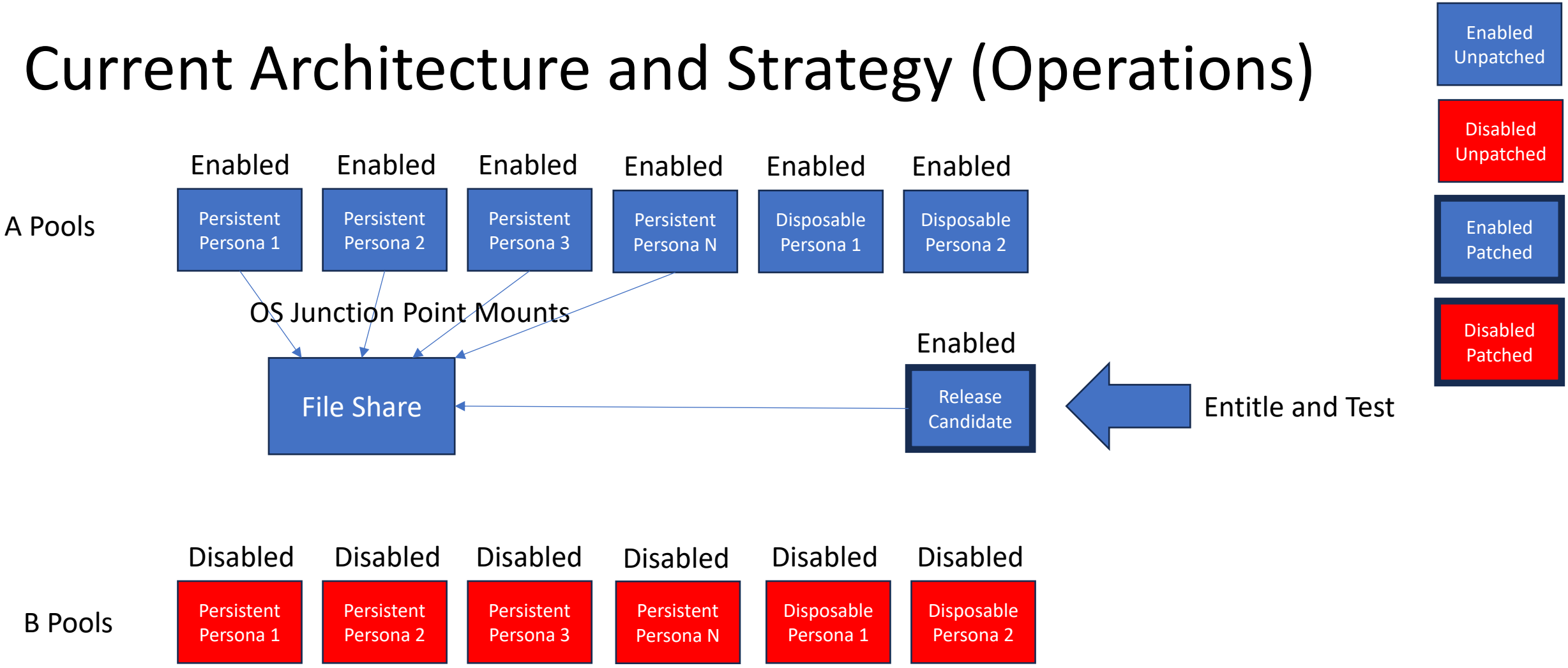

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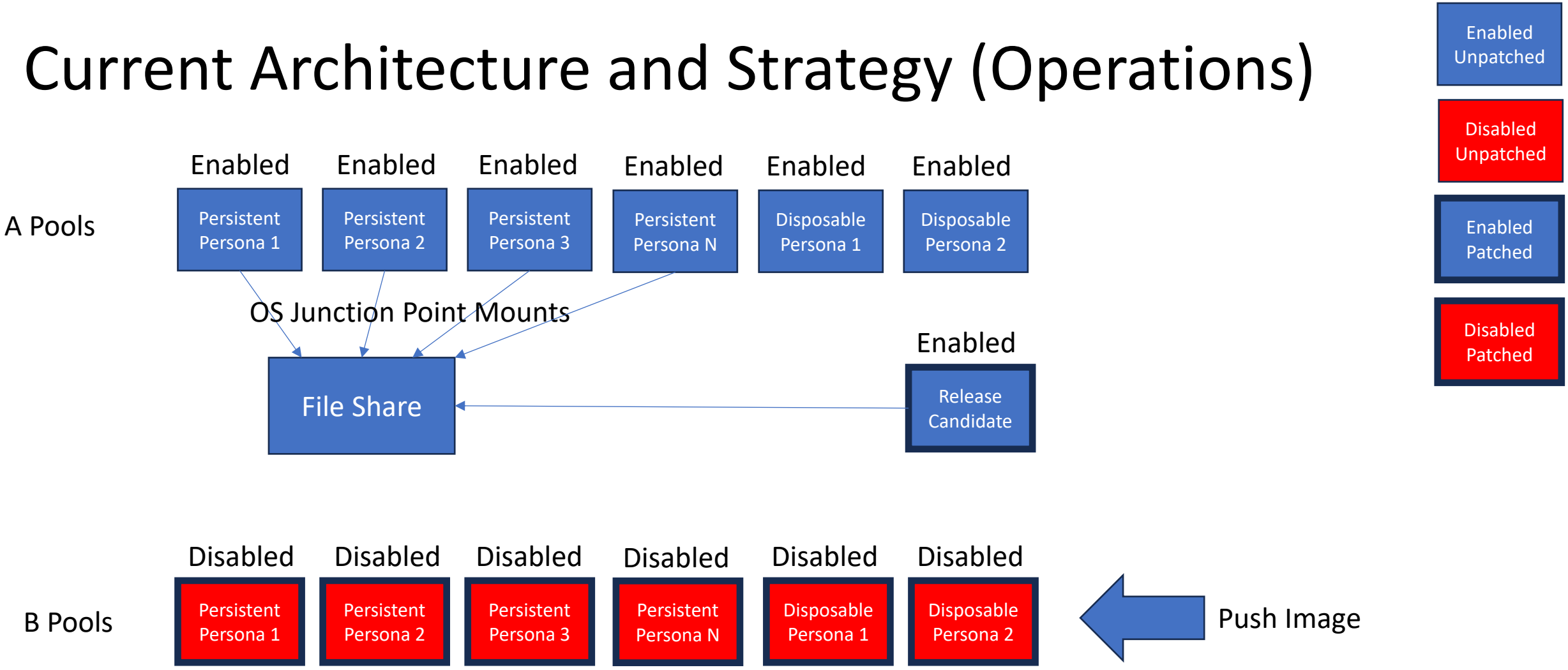

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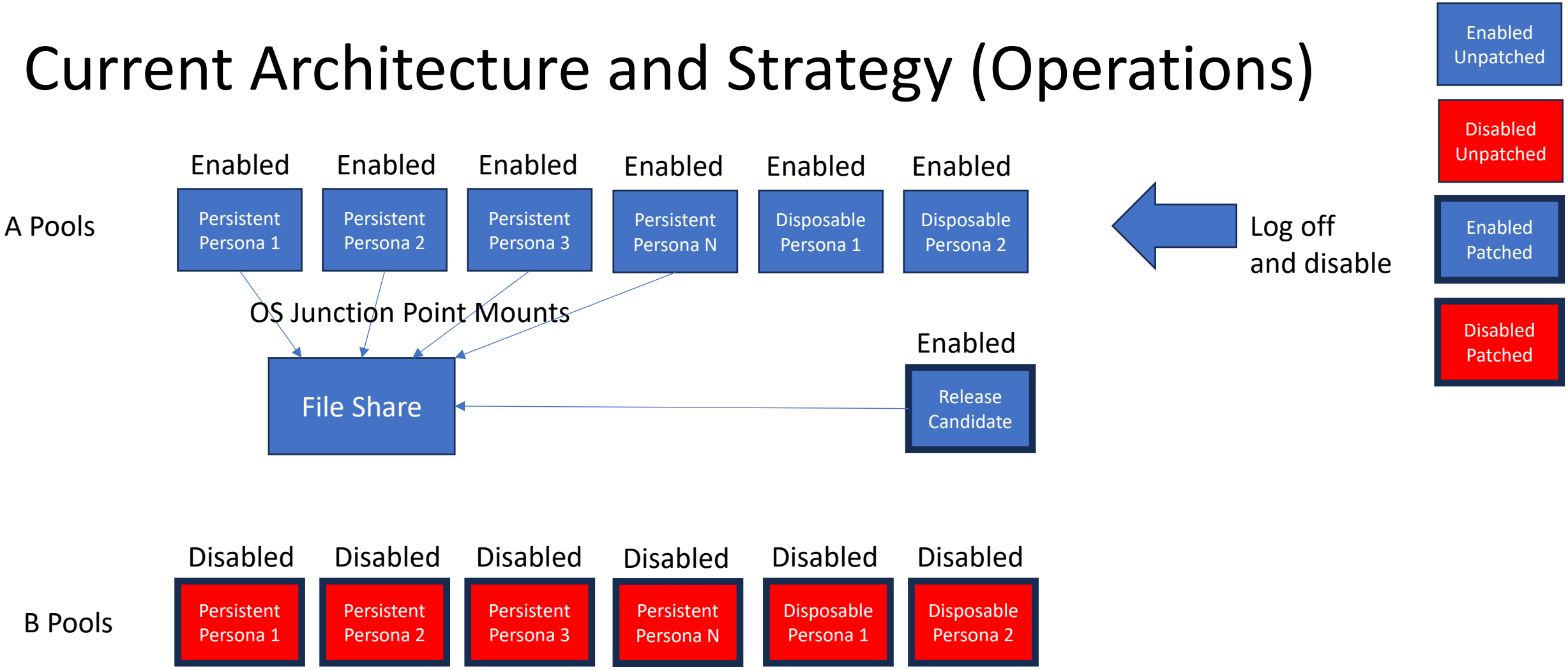

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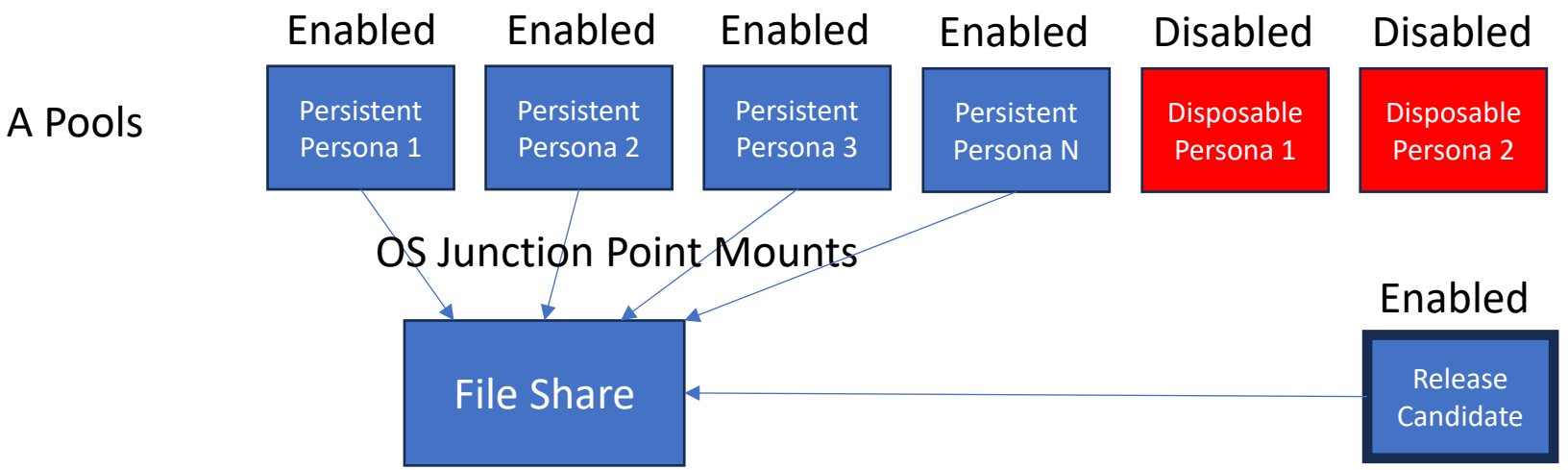
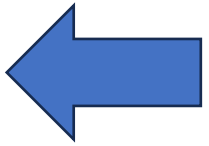
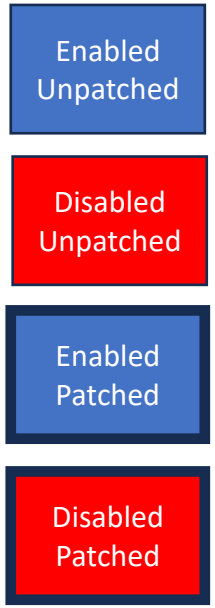




Enable Disposable Pools when users are out



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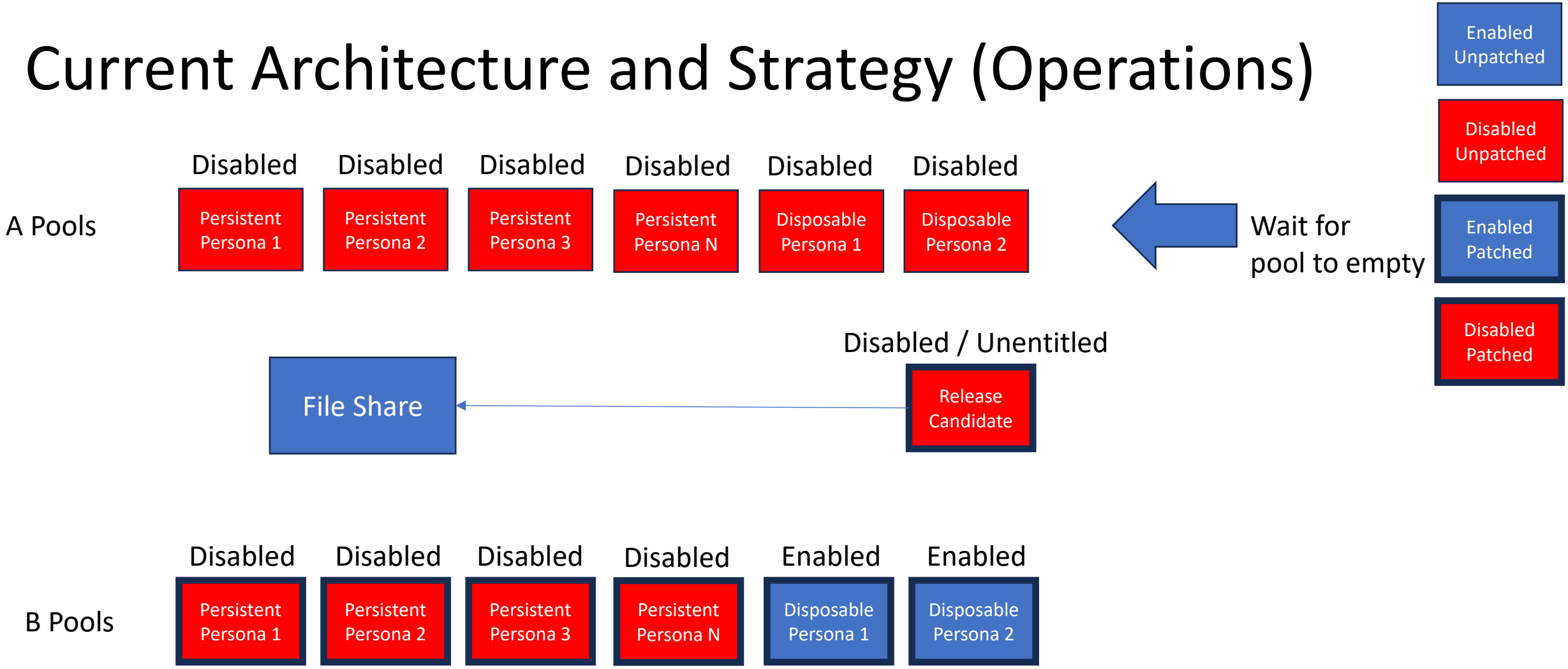

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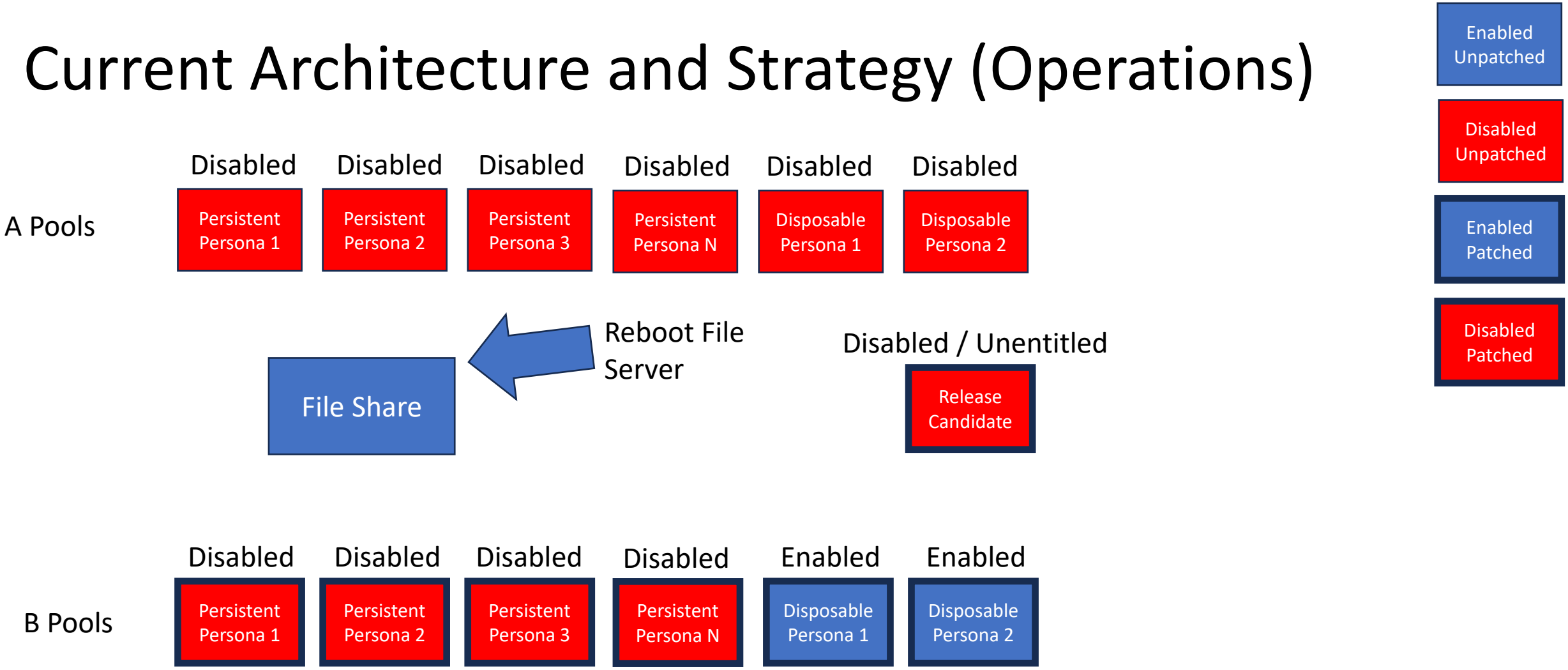

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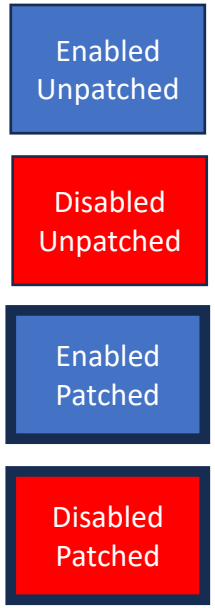


A Pools

Disabled	Disabled	Disabled	Disabled	Disabled	Disabled
Persistent Persona 1	Persistent Persona 2	Persistent Persona 3	Persistent Persona N	Disposable Persona 1	Disposable Persona 2

Disabled / Unentitled

Release Candidate

File Share

OS Junction Point Mounts

B Pools

Enabled	Enabled	Enabled	Enabled	Enabled	Enabled
Persistent Persona 1	Persistent Persona 2	Persistent Persona 3	Persistent Persona N	Disposable Persona 1	Disposable Persona 2

Enable Persistent Pools after File Share is up

Opportunities for Automation

- Challenge 1 – Spare Machines
  - Whatever the minimum desktop setting is for the disabled pools, those desktops will still be powered on
  - Need to Disable Provisioning on those pools and delete all VMs in the pool

- Challenge 2 – OS Upgrades
  - When upgrading the operating system you cannot simply point to the upgraded OS
    - Active Directory OU will stay the same and the new OS may have different GPO requirements
    - The new OS might have different TPM requirements (Windows 10 vs Windows 11)
  - Need to replace either the A or B side with the new OS and clone those pools

Opportunities for Automation – other tasks

- Challenge 3 – Send Message Functionality in Horizon
  - Often it is a good idea to notify connected users that maintenance is coming
  - The Horizon Administrator UI only lets you send messages to 100 selected users
  - Need to be able to send messages to all connected users

- Challenge 4 – Pool Backups
  - There isn’t a built-in way to back up a pool’s settings
  - Need to be able to back up all pool settings to JSON based configuration files

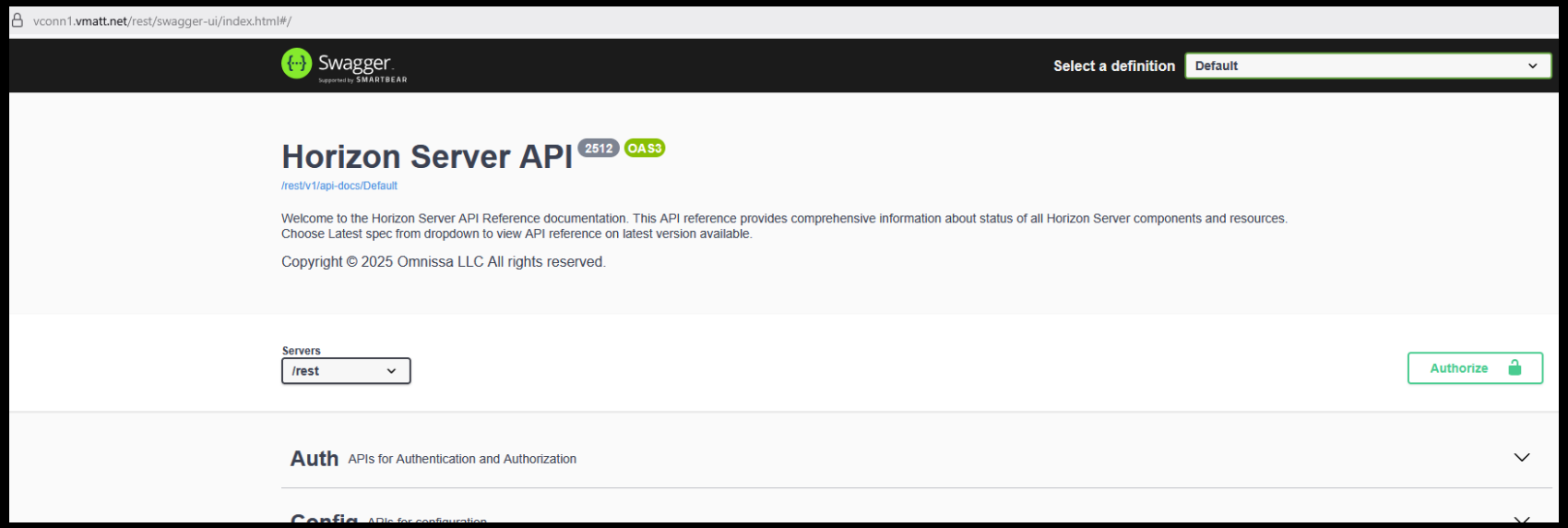
- Challenge 5 – Pushing an image to a pool

Horizon®

Using the REST API

Omnissa Horizon REST API

- Replaces the legacy Horizon SOAP API – Going away in Q2 2026
- Introduced in 2019 with Horizon 7.10
- Full Parity with SOAP API as of Horizon 8 Build 2312 (December 2023)
- Check it out at https://yourConnSvrFQDN/rest/swagger-ui/index.html



Omnissa Horizon REST API – Using Swagger

Auth APIs for Authentication and Authorization

POST `/key-agreement` Uses Diffie-Hellman algorithm to achieve encryption key agreement between client and server, which then can be used to encrypt and decrypt sensitive information.

GET `/login` Logs in a user with smart card. Returns access and refresh tokens.

POST `/login` Logs in a user. Returns access token and refresh token.

Note: UPN(e.g. testadmin@example.com) based login is not supported.

Parameters Try it out

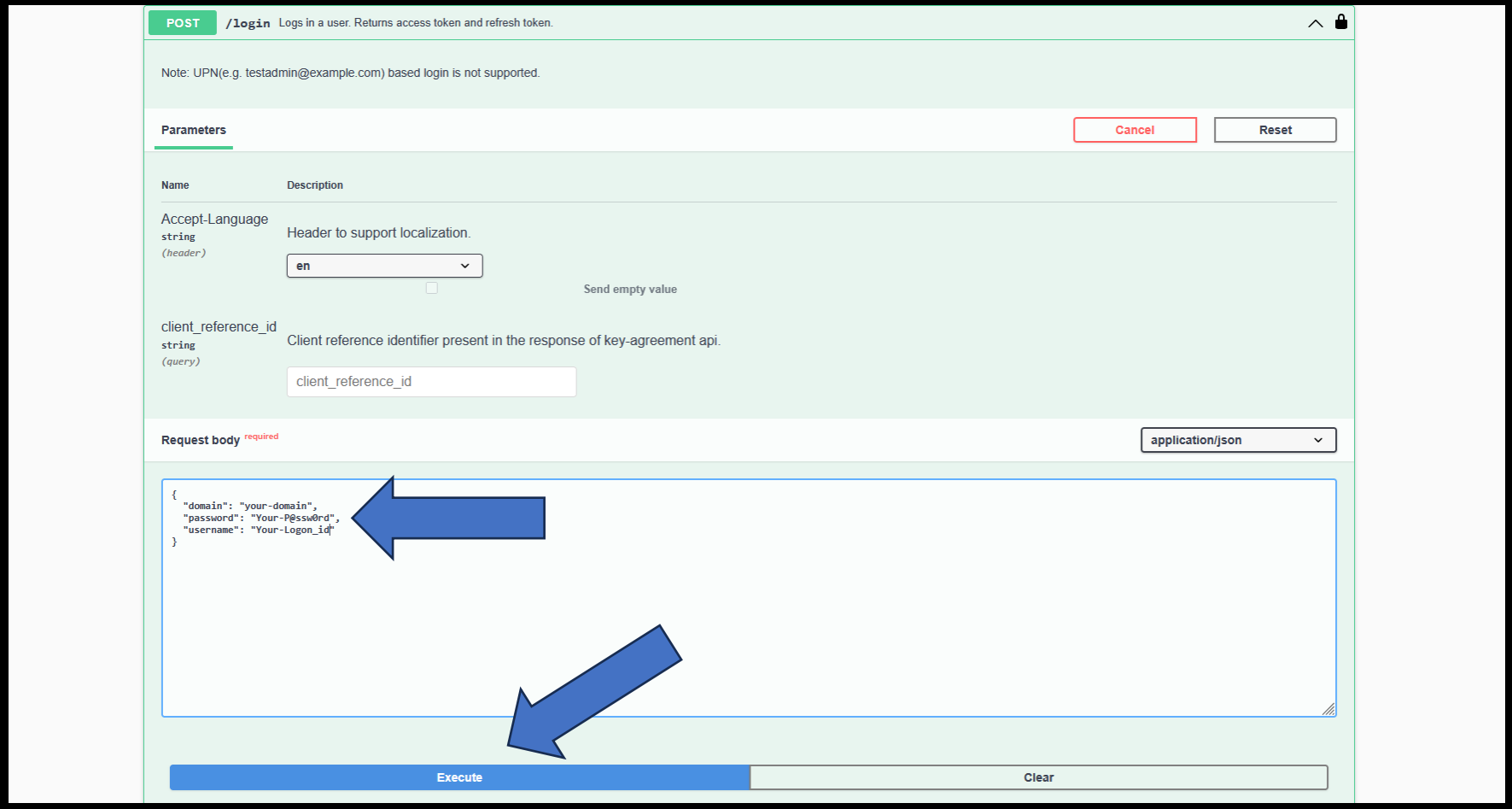
Name	Description
Accept-Language string *(header)*	Header to support localization. *Available values* : de, en, es, fr, ja, ko, zh-CN, zh-TW *Default value* : en en
client_reference_id string *(query)*	Client reference identifier present in the response of key-agreement api. client_reference_id

Request body required — application/json

Example Value | Schema

```
{
  "domain": "AD-TEST-DOMAIN",
  "password": "<password>",
  "username": "Administrator"
}
```

Omnissa Horizon REST API – Using Swagger



Omnissa Horizon REST API – Using Swagger

Responses

Curl

```
curl -X 'POST' \
  'https://[redacted]t/rest/login' \
  -H 'accept: */*' \
  -H 'Accept-Language: en' \
  -H 'Content-Type: application/json' \
  -d '{
  "domain": "vm[redacted]",
  "password": "[redacted]",
  "username": "[redacted]
}'
```

Request URL

```
https://v[redacted]et/rest/login
```

Server response

Code	Details
200	Response body


```
{
  "access_token": "eyJhbGciOiJSUzI1NiJ9.eyJ1c2VyLXNpZCI6I1MtMS01LTIxLTE1NTQ2ODk5NDktMTk2ODg5MjM5MC0yNjk2MjAyMDM0LTExMDUiLCJjbG91ZEFkbWluIjpmYWxzZSwiYnJva2VyIG1kIjoiVkNPTk4xIiwidG9rZW4gdHlwZSI6Im
FjY2VzcyB0b2tlbiIsInN1Ymp1Y3QiOiJtaGFkbWluIiwiZG9tYWluIjoidm1hdHQubmV0IiwiZG9tYWluLXRydXN0LXR5cGUiOiJQUk1NQVJZX0RPTUFJTiIsInVzZXItbG9jYWxlIjoiZW4iLCJzZXNzaW9uLWlkIjoiZGQ4MDkyN2YtYjAxOC00Y2NmLTg4
YjYtZjQwZGZjZDBiZGM1Iiwic3ViIjoibWhhZG1pbiIsImlhdCI6MTc3NTA2MDMyNywianRpIjoiZDkxOWE3Y2EtYzIyYS00NjY4LWIwYWItYTk2MmJjNjc3MDU0IiwibmJmIjoxNzc1MDYwMzI3LCJleHAiOjE3NzUwNjIxMjd9.pcLwwte7CzfCTFksdmtXW
7g0W1T1Q5cBDKD15GuOjafVzsN7WmJPLyaNxxu9fM1YDsrAvbRUxDFXT3U1zO3x3F_9GJzRDhCKQm_HhRlvidiHKQW2P4M5Sm-ngdTmmLqFI3Du4TAXo7aSw6gy1ix4JAZJ7Na-DI9d_8hAn1sBKuhuhqDby3tbw8kiKEBHwKsqYbg1JKsRqeXXm_2wsBW2kN0
ba8XTOEGKdY1-UNS2gUnnfohjKsDvlyT-QznG-jBQBkof3tkcg0z0BEYVfejd37GN4Ih9bgDbWm-68pshvbTQ1muAdCNbHd5lhRHscWE5wGdkjB5caMK4ReW8AkM0oA",
  "refresh_token": "eyJhbGciOiJSUzI1NiJ9.eyJ1c2VyLXNpZCI6I1MtMS01LTIxLTE1NTQ2ODk5NDktMTk2ODg5MjM5MC0yNjk2MjAyMDM0LTExMDUiLCJjbG91ZEFkbWluIjpmYWxzZSwiYnJva2VyIG1kIjoiVkNPTk4xIiwidG9rZW4gdHlwZSI6I
nJlZnJlc2ggdG9rZW4iLCJzdWJqZWN0IjoibWhhZG1pbiIsImRvbWFpbiI6InZtYXR0Lm5ldCIsImRvbWFpbi10cnVzdC10eXB1IjoiUFJJTUFSWV9ET01BSU4iLCJ1c2VyLWxvY2FsZSI6ImVuIiwic2Vzc2lvbi1pZCI6ImRkODA5MjdmLWIwMTgtNGNjZi0
4OGI2LWY0MGRmY2QwYmRjNSIsInN1YiI6Im1oYWRtaW4iLCJpYXQiOjE3NzUwNjAzMjcsImp0aSI6IjcxY2UzM2Q2LWY3YTUtNDViOC04MzQzLTliZWE5YWU4YzU0YyIsIm5iZiI6MTc3NTA2MDMyNywiZXhwIjoxNzc1MDYyMTI3fQ.CBkjaUDX821uw5O847
8NysZHzy9IO9mmHmSGXH6GKa0cGTtWUDC3bupu1kcVVMYQf4E0Zlp2JcbwF45-YRwJy11HJQ2xygniMc6CCBH413tO9wImsYLc6yrocbBfwd2-IuU6e-TXa0Rmhj6wBXDXQThOqbDrsMnuuptRjgZnihPYyGolqxMs0YvnZvq-DA3ks5
5uR2znLz-Sz50noZd87RAZ761iBYnaxh2g1ndUpf1D1FXXtB6XVhyTn1hhp9UkMxpINf4gGfvanX6jjtGlqh2hZWshEjVyzeMR461Cj16cHz4A434aVaBVbLKpaZP5fO5w"
}
```


Download

Response headers

```
access-control-allow-credentials: true
access-control-allow-origin: https:/[redacted]
content-security-policy: default-src 'none';connect-src 'self' https:;font-src 'self' data:;img-src 'self' data:;base-uri 'none';form-action 'none';frame-ancestors 'none';script-src
'self';style-src 'self' 'unsafe-inline'
content-type: application/json
referrer-policy: strict-origin-when-cross-origin
strict-transport-security: max-age=31536000
transfer-encoding: chunked
vary: Origin
x-content-type-options: nosniff
x-permitted-cross-domain-policies: none
x-robots-tag: none
```

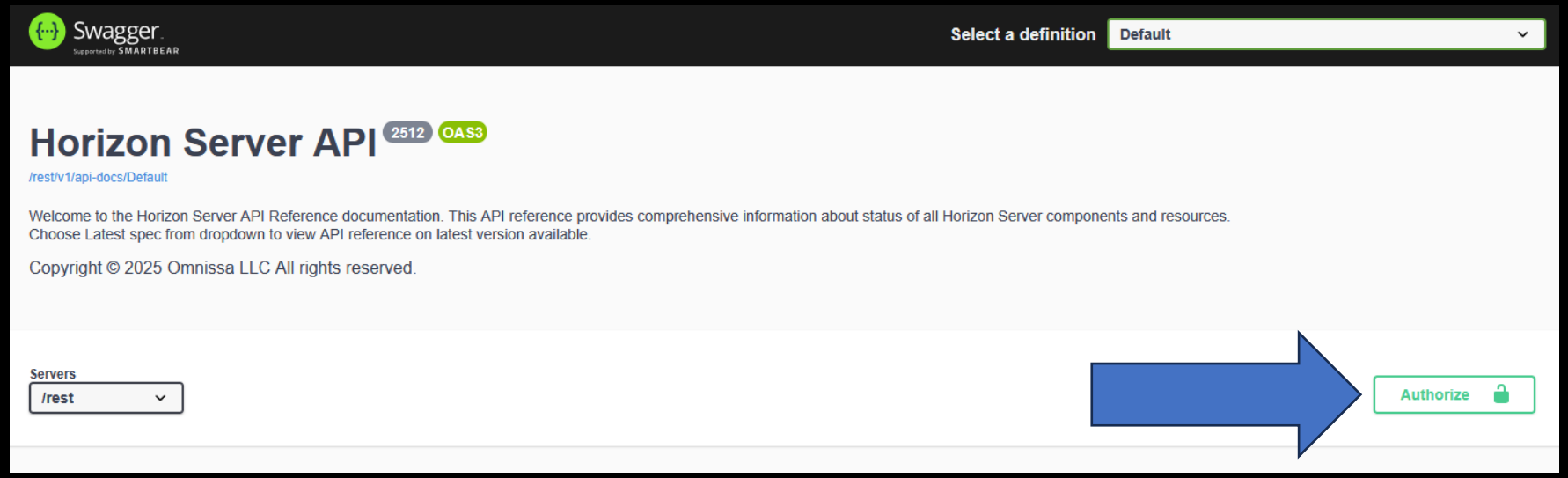
Responses

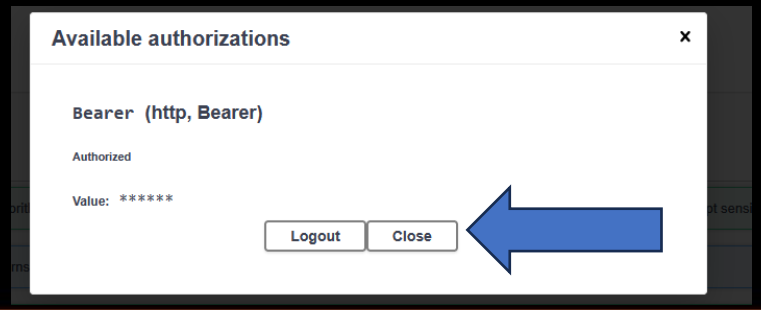
Code	Description	Links
200	OK	*No links*

Grab access token (inside quotes)

Omnissa Horizon REST API – Using Swagger

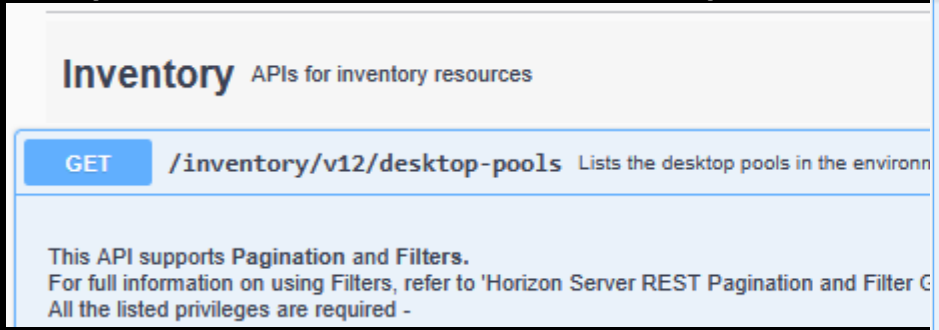
Scroll to the top and click “authorize” – Sessions are typically 30 minutes



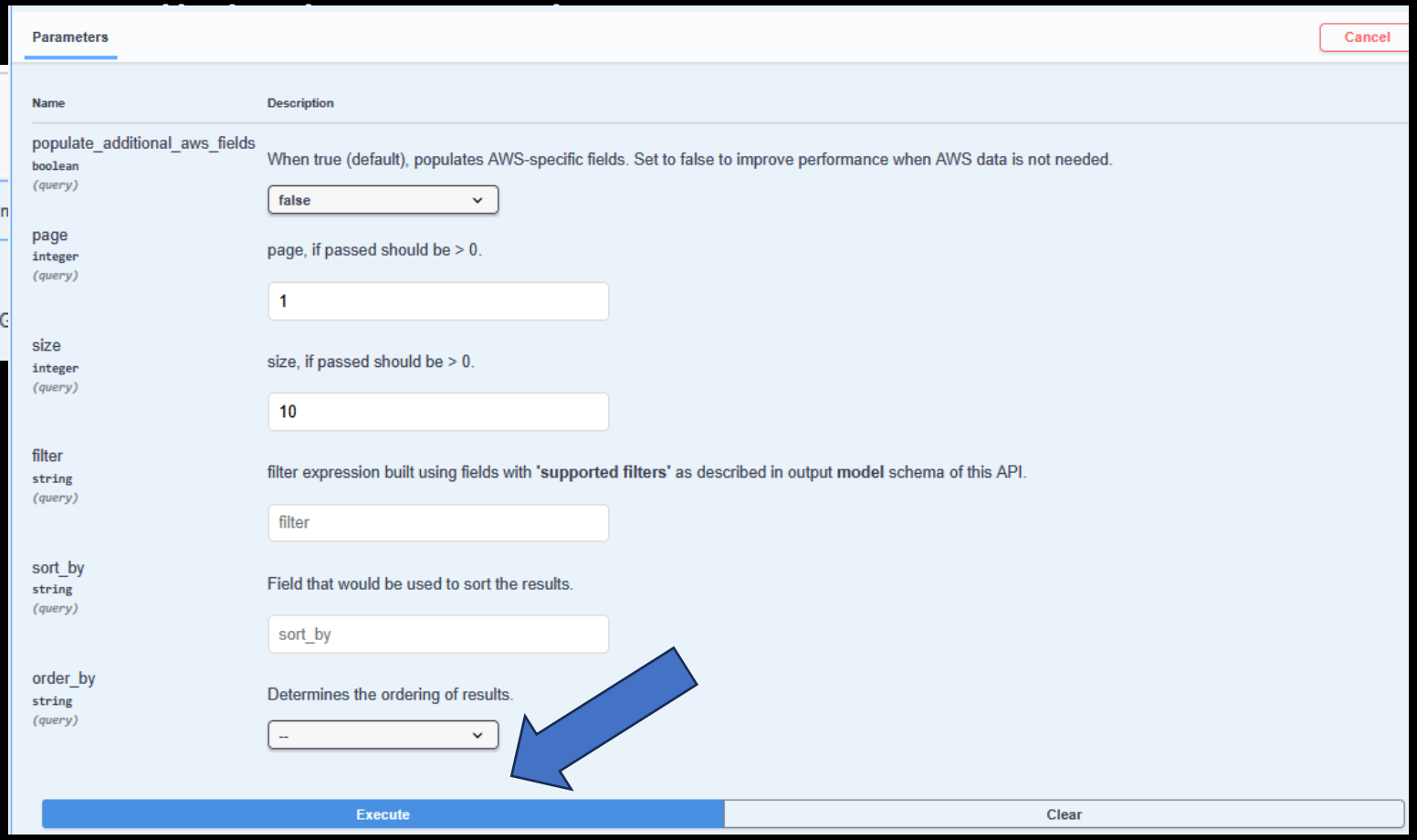


Omnissa Horizon REST API – Using Swagger

Try out a different endpoint –



Clear the values in the 1st, 4th, 5th and 6th parameters and click EXECUTE



Omnissa Horizon REST API – Using Swagger

Responses

Curl

```
curl -X 'GET' \
  'https://vconn1.vmatt.net/rest/inventory/v12/desktop-pools?populate_additional_aws_fields=false&page=1&size=10' \
  -H 'accept: */*' \
  -H 'Authorization: Bearer eyJhbGciOiJSUzI1NiJ9.eyJ1c2VyLXNpZCI6IlMtMS01LTIxLTE1NTQ2ODk5NDktMTk2ODg5MjM5MC0yNjk2MjAyMDM0LTExMDUiLCJjbG91ZEFkbWluIjpmYWxzZSwiYnJva2VyIGlkIjoiVkNPTk4xIiwidG9rZW4gdHlwZSI6Im
```

Request URL

```
https://vconn1.vmatt.net/rest/inventory/v12/desktop-pools?populate_additional_aws_fields=false&page=1&size=10
```

Server response

Code	Details
200	Response body


```
[
  {
    "id": "2140a164-1f31-4143-b6f7-c61619f2a9f5",
    "name": "WIN11-FC1",
    "display_name": "Windows 11 FC1",
    "access_group_id": "3ae27f8c-cc49-4273-b08d-b1467f9428bb",
    "enabled": true,
    "delete_in_progress": false,
    "enable_client_restrictions": false,
    "session_type": "DESKTOP",
    "cloud_managed": false,
    "cloud_assigned": false,
    "display_assigned_machine_name": false,
    "type": "MANUAL",
    "source": "VIRTUAL_CENTER",
    "vcenter_id": "be755822-7b58-46bc-a570-f12394e7fb6b",
    "user_assignment": "DEDICATED",
    "automatic_user_assignment": true,
    "allow_multiple_user_assignments": false,
    "transparent_page_sharing_scope": "VM",
    "session_settings": {
      "power_policy": "TAKE_NO_POWER_ACTION",
      "disconnected_session_timeout_policy": "NEVER",
      "allow_users_to_reset_machines": true,
      "allow_multiple_sessions_per_user": false
    },
    "display_protocol_settings": {
      "display_protocols": [
        "PCoIP"
```


Download

Response headers

```
cache-control: no-cache,no-store,max-age=0,must-revalidate
content-security-policy: default-src 'none';connect-src 'self' https:;font-src 'self' data:;img-src 'self' data:;base-uri 'none';form-action 'none';frame-ancestors 'none';script-src
'self';style-src 'self' 'unsafe-inline'
content-type: application/json
expires: 0
pragma: no-cache
referrer-policy: strict-origin-when-cross-origin
strict-transport-security: max-age=31536000 ; includeSubDomains
transfer-encoding: chunked
x-content-type-options: nosniff
x-frame-options: DENY
x-permitted-cross-domain-policies: none
x-robots-tag: none
x-total-count: 5
x-xss-protection: 0
```

Omnissa Horizon REST API – Using Swagger

Paste the Pool ID from the “all pools” result and choose a new endpoint

GET /inventory/v12/desktop-pools/{id} Gets the desktop pool information.

All the listed privileges are required -

Privileges	Description
POOL_VIEW	Caller should have permission to this privilege, on the associated access group of the desktop pool.
FEDERATED_LDAP_VIEW	Caller should have permission to this privilege, on the associated federation access group, to read the global desktop entitlement ID field of a desktop pool. Else, it will be unset.

Parameters

Cancel

Name	Description
id * required string (path)	23d72420-4e18-488c-8f46-caac1ba2620b

Execute

Omnissa Horizon REST API – Using Swagger

Responses

Curl

```
curl -X 'GET' \
  'https://vconn1.vmatt.net/rest/inventory/v12/desktop-pools/23d72420-4e18-488c-8f46-caac1ba2620b' \
  -H 'accept: */*' \
  -H 'Authorization: Bearer eyJhbGciOiJSUzI1NiJ9.eyJ1c2VyLXNpZCI6IlMtMS01LTIxLTE1NTQ2ODk5NDktMTk2ODg5MjM5MC0yNjk2MjAyMDM0LTExMDUiLCJjbG91ZEFkbWluIjpmYWxzZSwiYnJva2VyIGlkIjoiVkNPTk4xIiwidG9rZW4gdHlwZSI6Im
```

Request URL

```
https://vconn1.vmatt.net/rest/inventory/v12/desktop-pools/23d72420-4e18-488c-8f46-caac1ba2620b
```

Server response

Code	Details
200	Response body


```
    "number_of_spare_machines": 6
  },
  "provisioning_settings": {
    "min_ready_vms_on_vcomposer_maintenance": 0,
    "add_virtual_tpm": false,
    "parent_vm_id": "vm-21032",
    "base_snapshot_id": "snapshot-21284",
    "datacenter_id": "datacenter-1001",
    "vm_folder_id": "group-v1002",
    "host_or_cluster_id": "domain-c18004",
    "resource_pool_id": "resgroup-18005",
    "compute_profile_num_cores_per_socket": 4,
    "compute_profile_num_cpus": 4,
    "compute_profile_ram_mb": 8172,
    "resource_pool_path": "/VMATT/host/CL-HP1/Resources",
    "snapshot_path": "/2025-12-09 - post migrate to VCF/202601081A/202601221A/post-DVS/20260327_1",
    "parent_vm_path": "/VMATT/vm/W11-23H2-1",
    "vm_folder_path": "/VMATT/vm/WIN2A",
    "datacenter_name": "VMATT",
    "host_or_cluster_path": "/VMATT/host/CL-HP1"
  },
  "storage_settings": {
    "datastores": [
      {
        "datastore_id": "datastore-15006",
        "storage_overcommit": "UNBOUNDED"
      }
    ],
    "use_vsan": false,
```


Download

Response headers

```
cache-control: no-cache,no-store,max-age=0,must-revalidate
content-security-policy: default-src 'none';connect-src 'self' https:;font-src 'self' data:;img-src 'self' data:;base-uri 'none';form-action 'none';frame-ancestors 'none';script-src
'self';style-src 'self' 'unsafe-inline'
content-type: application/json
expires: 0
pragma: no-cache
referrer-policy: strict-origin-when-cross-origin
strict-transport-security: max-age=31536000 ; includeSubDomains
transfer-encoding: chunked
x-content-type-options: nosniff
x-frame-options: DENY
x-permitted-cross-domain-policies: none
x-robots-tag: none
x-xss-protection: 0
```

Using Powershell

Basic Functions for your scripts

- Get-TokenExpiry
  - Looks to see if $token exists
    - If it does, it gets the expiration time
    - If not, it skips the check
  - Extracts the expiration time from token
  - Returns an object $expirationTime
  - Value is in UTC

```
# Function to check for a valid Horizon REST API token
14 references
function Get-TokenExpiry {

    # If token is null, empty, or whitespace — skip the check entirely
    if ([string]::IsNullOrWhiteSpace($token)) {
        Write-Host "No token found — skipping expiration check." -ForegroundColor Yellow
        return $null
    }

    try {
        $tokenParts = $token.Split('.')

        $decodedPayload = [System.Text.Encoding]::UTF8.GetString(
            [Convert]::FromBase64String($tokenParts[1])
        )

        $decodedPayloadObj = $decodedPayload | ConvertFrom-Json
        $expirationTimestamp = $decodedPayloadObj.exp

        $expirationTime = [DateTimeOffset]::FromUnixTimeSeconds($expirationTimestamp).UtcDateTime

        Write-Host "Token expires at: $expirationTime"
        return $expirationTime
    }
    catch {
        Write-Host "Token is invalid or unreadable — skipping expiration check." -ForegroundColor Yellow
        return $null
    }
}
```

Basic Functions for your scripts

- Get-HorizonToken
  - Prompts for credentials
  - Executes a POST to the /rest/login API endpoint
  - Returns an object $token
  - Requires $connectionServer to be valid in your script

Important Security Note

Source: *PowerShell Security Best Practices (Microsoft Learn)*
https://learn.microsoft.com/powershell/scripting/security/secure-strings

Microsoft explicitly states:

"Use Get-Credential *to prompt the user for credentials. This cmdlet returns a PSCredential object that contains a SecureString password."*

And:

"If you must use a plaintext password—for example, to call an API—convert the SecureString only at the point of use."

```powershell
function Get-HorizonToken {

    Write-Host "Token invalid - Re-authenticating" -ForegroundColor Yellow

    # Prompt for Horizon admin credentials
    $creds = Get-Credential -Message "Enter Horizon admin credentials"

    # Extract username + password
    $username = $creds.UserName
    $password = $creds.GetNetworkCredential().Password

    # If user typed DOMAIN\username, split it
    if ($username -match "\\") {
        $domain, $username = $username -split "\\", 2
    }

    # Build authentication URI
    $authUri = "$connectionServer/rest/login"

    # Build JSON body
    $body = @{
        username = $username
        password = $password
        domain   = $domain
    } | ConvertTo-Json

    # Send authentication request
    $response = Invoke-RestMethod -Method Post -Uri $authUri `
        -Body $body `
        -ContentType "application/json"
        # -SkipCertificateCheck

    # Extract token
    $token = $response.access_token

    Write-Host "Authenticating to " $authUri

    return $token
}
```

Basic Functions for your scripts

Combine the two functions in the body of your scripts

```
Write-Host "Current UTC Time:" (Get-Date).ToUniversalTime().ToString("MM/dd/yyyy HH:mm:ss")
$tokenExpirationTime = Get-TokenExpiry


if ($tokenExpirationTime -gt (Get-Date -AsUTC)) {
    $tokenStatus = $true
} else {
    $token = Get-HorizonToken
    $tokenExpirationTime = Get-TokenExpiry
}
```

Basic Functions for your scripts

- Get-HorizonVersion
  - Hits the Horizon REST API documentation at /rest/v1/api-docs/Default
  - Searches
  - Returns an object $version
  - Helps when pointing your server URI to the proper endpoint
  - $horizonVersion = Get-HorizonVersion
  - $connectionServer needs to exist

```
Function Get-HorizonVersion {

$apiDocsUri = "$connectionServer/rest/v1/api-docs/Default"

$headers = @{
    Authorization = "Bearer $token"
  }

  $apiDocs = Invoke-RestMethod -Method GET -Uri $apiDocsUri -Headers $headers
$version = $apiDocs.info.version
return $version
}
```

Raw Data within api-docs


```
{
  "openapi": "3.0.1",
  "info": {
    "description": "Welcome to the Horizon Server API Reference
documentation. This API reference provides comprehensive
information about status of all Horizon Server components and
resources. \u003Cbr\u003E \nChoose Latest spec from dropdown to
view API reference on latest version available.",
    "license": {
      "name": "Copyright © 2025 Omnissa LLC All rights reserved."
    },
    "title": "Horizon Server API",
    "version": "2512"
  },
```

Basic Functions for your scripts

- Using Get-HorizonVersion
  - Validate token
  - Get Horizon Version
  - Desktop Pools endpoint
    - Use switch command to establish newest desktop pools API endpoint version
    - Needs to be updated manually
    - Get all pools example below

```
# Get all pools
$poolsUri = "$connectionServer/rest/inventory/$dpVer/desktop-pools"
$pools = Invoke-RestMethod -Method GET -Uri $poolsUri -Headers @{
    "Authorization" = "Bearer $token"
    "Accept"        = "application/json"
} # -SkipCertificateCheck
```

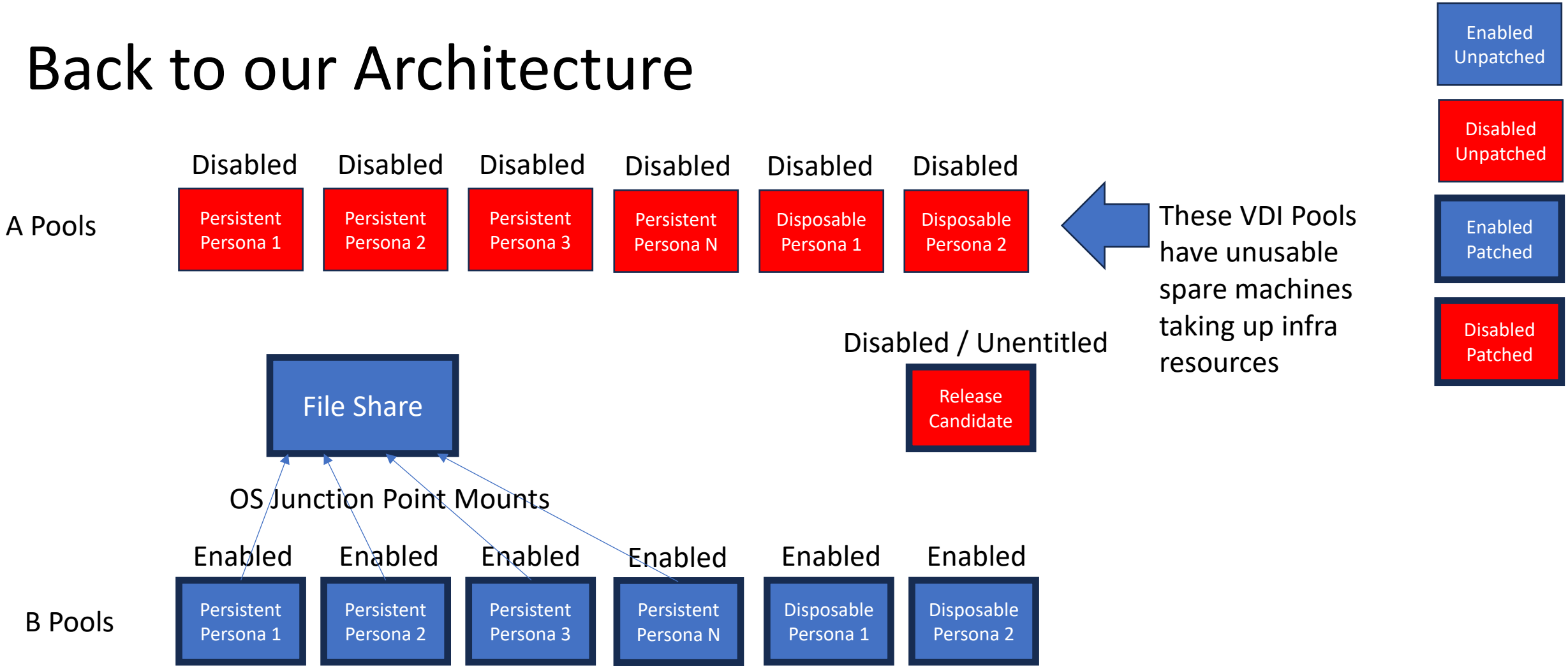
```
Write-Host "Current UTC Time:" (Get-Date).ToUniversalTime().ToString("MM/dd/yyyy HH:mm:ss")
$tokenExpirationTime = Get-TokenExpiry

$horizonVersion = Get-HorizonVersion
Write-host $horizonVersion

if ($tokenExpirationTime -gt (Get-Date -AsUTC)) {
    $tokenStatus = $true
} else {
    $token = Get-HorizonToken
    $tokenExpirationTime = Get-TokenExpiry
}

# Desktop Pools - Key API Endpoint Versions
switch ($horizonVersion) {
    "2512" { $dpVer = "v12" }
    "2506" { $dpVer = "v11" }
    "2503" { $dpVer = "v10" }
    "2412" { $dpVer = "v10" }
    "2406" { $dpVer = "v8"  }
    "2312" { $dpVer = "v8"  }
    "2309" { $dpVer = "v7"  }
    default { $dpVer = "v12" }
}
```

Back to our Architecture



Opportunities for Automation

- Challenge 1 – Spare Machines
  - Whatever the minimum desktop setting is for the disabled pools, those desktops will still be powered on
  - Need to Disable Provisioning on those pools and delete all VMs in the pool

Let’s fix it!

Opportunities for Automation – Remove Spares

- (Step 1) Modify Pool Settings
  - Set “poolNameToModify” at top of script along with connection server and domain
  - Get all pools
  - Find the pool ID associated with $poolNameToModify
  - Extract all pool settings for that specific pool to an object $updated
  - Ask the user if they would like to disable provisioning and/or the pool itself
  - Extract the values of the answers and make a Boolean
  - Change the values to $updated.enable_provisioning and $updated.enabled
  - Convert $updated to a JSON
  - PUT the JSON to “/rest/inventory/$dpver/desktop-pools$poolId”

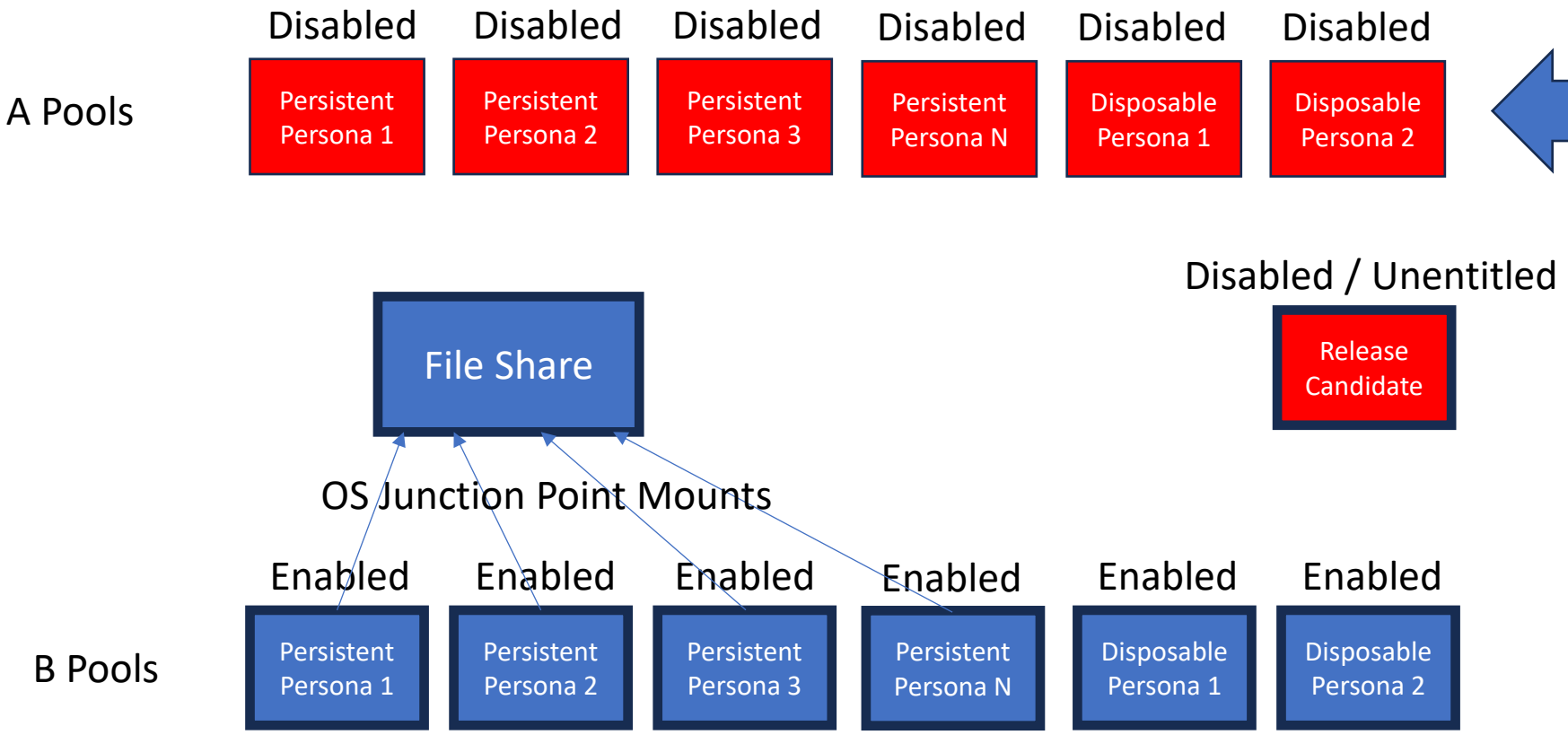
Opportunities for Automation – Remove Spares

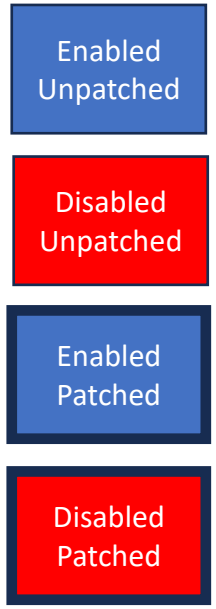
- (Step 2) Delete all machines in a pool
  - Get a list of all machines
  - Find machines where the pool ID matches and the state is AVAILABLE
  - If machines > 0
    - Build a list of machine ID's
    - Form JSON to delete from disk and not force logoff
  - If machines not > 0
    - Inform the user there are no machines found
- NOTE: If you left provisioning on in Step 1, machines will be recreated

Opportunities for Automation – Remove Spares

DEMO

What about Operating System Upgrades?





Opportunities for Automation

- Challenge 2 – OS Upgrades
  - When upgrading the operating system you cannot always point to the upgraded OS
    - Active Directory OU will stay the same and the new OS may have different GPO requirements
    - The new OS might have different TPM requirements (Windows 10 vs Windows 11)
  - Need to replace either the A or B side with the new OS and clone those pools

Let’s fix it!

What about Operating System Updates?

A Pools

Enabled | Enabled | Enabled | Enabled | Enabled | Enabled

NEW OS | NEW OS | NEW OS | NEW OS | NEW OS | NEW OS

OS Junction Point Mounts

File Share

Disabled / Unentitled

NEW OS

Disabled / Unentitled

Release Candidate

B Pools

Old pools are deleted after cutover. How do we clone the A pools to the B pools?

Opportunities for Automation

- Challenge 2 – Clone Desktop Pool to a new Pool - TO DO LIST
  - Run functions to Validate / Create API Token, determine Horizon version
  - Get all pool information
  - Match pool name to pool ID and import that pool to a PS object $currentPool
  - Remove the “id” object so it gets recreated
  - Change the pool “name” and “naming pattern”
  - Ask user if they want to enable the pool and/or enable provisioning
  - Import changes and clone the pool with new properties
  - Ask user if they want to clone entitlements from original pool

Opportunities for Automation

- Challenge 2 – Clone Desktop Pool to a new Pool

DEMO

Opportunities for Automation

- Challenge 3 – Send Message Functionality in Horizon
  - Often it is a good idea to notify connected users that maintenance is coming
  - The Horizon Administrator UI only lets you send messages to 100 selected users
  - Need to be able to send messages to all connected users

Let’s fix it!

Horizon® 8: Cluster-VCONN1

vconn1.vmatt.net/admin/#/main/sessions_list 80%

Horizon 8 | Pod Cluster-VCONN1 | User Search | About | mhadmin

Updated 04/03/2026, 9:40 AM

- Sessions 4
- Problem vCenter VMs 0
- Problem Capacity Provider ... 0
- Problem RDS Hosts 0
- Events 2
- System Health 3

Monitor

- Dashboard
- Events
- Infrastructure
- Sessions
- Help Desk

Users and Groups

Sessions

Sessions: All

DISCONNECT SESSION | LOGOFF SESSION | RESTART DESKTOP | RESET VIRTUAL MACHINE | SEND MESSAGE

User	Type	Pool or Farm	Bypass Session Timeout	DNS Name	Client ID
vmatt.net\uo5221ps	Desktop	Windows Pool 1		win1a04.vmatt.net	cf50e77c2cd2d7d9c5
vmatt.net\uo5221ps	Desktop	Windows Pool 1		win1b01.vmatt.net	cf50e77c2cd2d7d9c5
vmatt.net\uo5221ps	Desktop	Windows Pool 2		win2b08.vmatt.net	[illegible]0e77c2cd2d7d9c5
vmatt.net\uo5221ps	Desktop	Windows Pool 2		win2a04.vmatt.net	[illegible]e77c2cd2d7d9c5

MANAGE COLUMNS

Rows per page 20 (20, 50, 100) | 1 - 4 of 4 rows

Opportunities for Automation

- Challenge 3 – Send Message Functionality in Horizon
  - Run functions to Validate / Create API Token
  - Get a list of all sessions and create an array $sessionIds
  - Ask user for the message to send, message type, and verification to send
  - Create $messagePayload JSON containing the message info and the array of sessions
  - POST JSON body to /rest/inventory/v1/sessions/action/send-message

Opportunities for Automation

- Challenge 4 – Pool Backups
  - There isn't a built-in way to back up a pool's settings
  - Need to be able to back up all pool settings to JSON based configuration files

Let's fix it!

Opportunities for Automation

- Challenge 4 – Pool Backups – TO DO
  - Establish Connection Server and Domain objects
  - Validate / create token / get horizon build version and assign $dpVer
  - Establish export path object
    - If path doesn’t exist, create the folder
  - GET all pools using /rest/inventory/$dpVer/desktop-pools endpoint
  - Iterate through all pools
    - Get pool ID
    - Save Pool settings to export folder using the pool name associated with the pool ID

Opportunities for Automation

- Challenge 5 – Pushing an image to a pool
  - Push Image is a manual process
  - Need to create a script that
    - Allows for a change of compute resources
    - Allows for a change of snapshot
    - Allows for a change of gold image VM name

Let’s fix it!

Thank You!

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